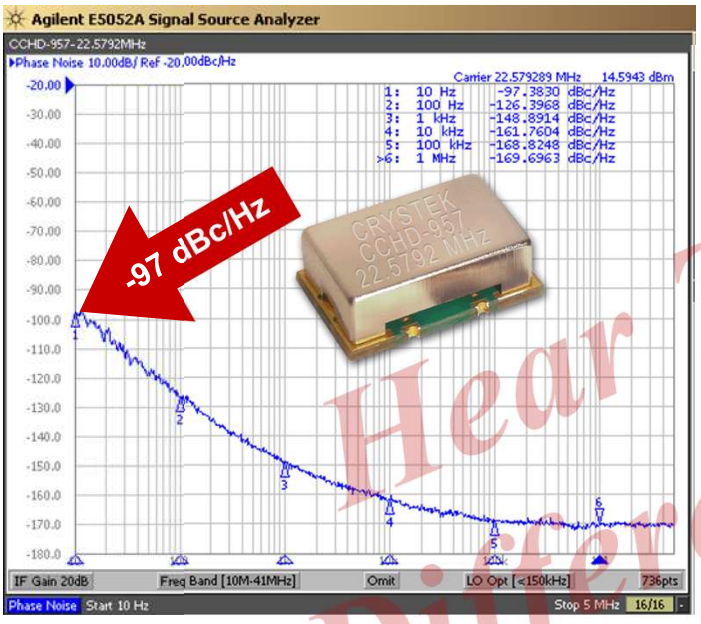
CCHD-957 FEMTO Clock Ultra-Low Phase Noise Oscillator with Standby Mode

CRYSTEK CRYSTALS
A DIVISION OF CRYSTEK CORPORATION

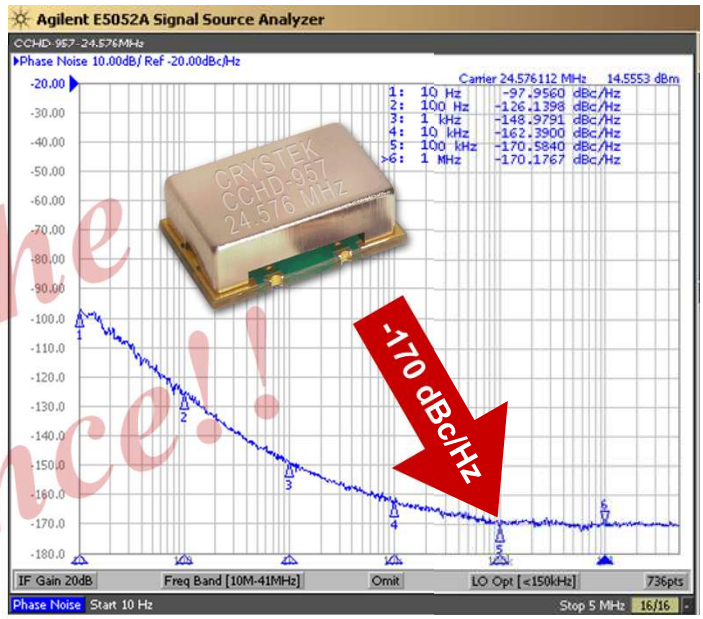
CCHD-957 Model
9×14 mm SMD, **3.3V, HCMOS**

RoHS Compliant

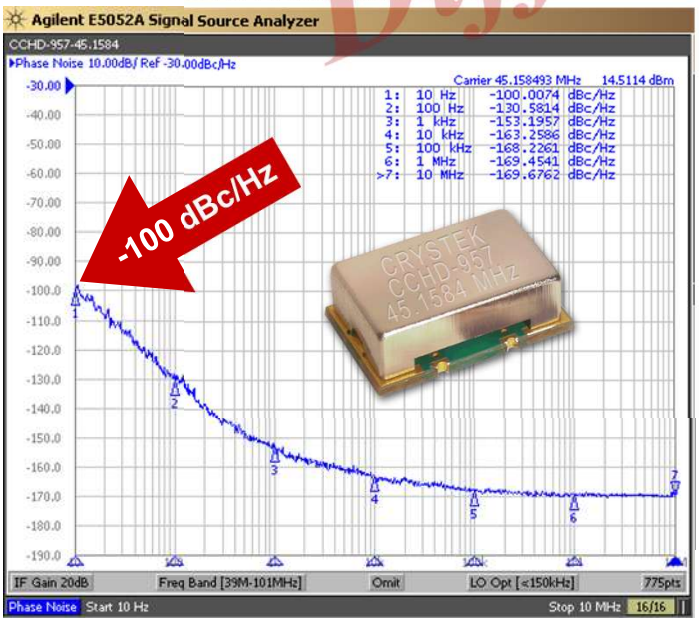
22.5792 MHz HCMOS 3.3V



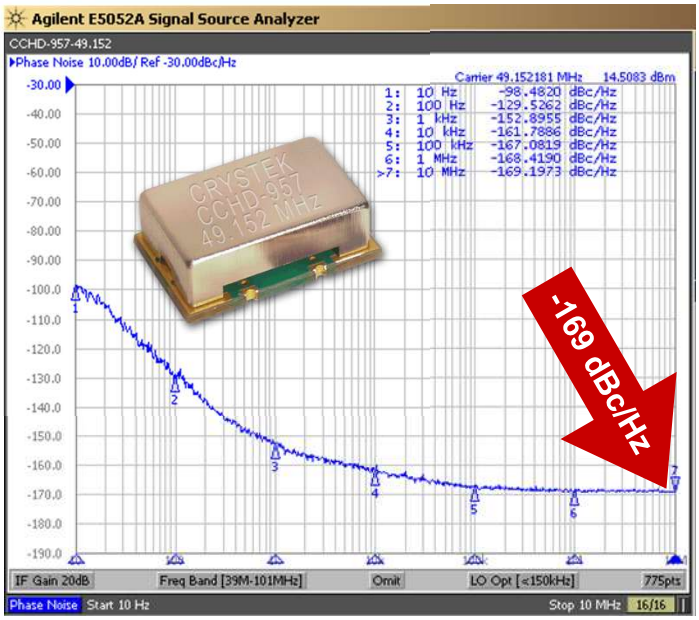
24.576 MHz HCMOS 3.3V



45.1584 MHz HCMOS 3.3V



49.152 MHz HCMOS 3.3V



Hear The Difference!!

Crystek's Model CCHD-957 HCMOS CLOCK oscillator family has been designed specifically for High Definition Audio (HD Audio). It features a typical low close-in phase noise of -100 dBc/Hz @ 10 Hz offset, and a noise floor of -169 dBc/Hz. With this extreme low phase noise performance, you will "Hear the Difference". It also features a "Standby Function", that is, when placed in disable mode, the internal oscillator is completely shut down in addition to its output buffer being placed in Tri-State. This family is housed in a 9×14 mm SMT package and operates with a +3.3V power supply.

Applications include:
Digital Audio Broadcasting (DAB)
Professional CD audio equipment
DACs and ADCs for HD audio

Rev: N
Date: 25-Aug-2026

CRYSTEK CORPORATION
16850 Oriole Road • Fort Myers, Florida 33912
Phone: 239-561-3311 • 800-237-3061
www.crystek.com

CCHD-957 FEMTO Clock Ultra-Low Phase Noise Oscillator with Standby Mode

CRYSTEK CRYSTALS
A Division of Crystek Corporation

CCHD-957 Model
9×14 mm SMD, **3.3V, HCMOS**

RoHS Compliant

Frequency Range:	**10 MHz to 50 MHz**
Temperature Range:	**0°C to +70°C**
(Option M)	**-20°C to +70°C**
(Option X)	**-40°C to +85°C**
Storage:	**-45°C to 90°C**
Input Voltage:	**3.3V ±0.3V**
Input Current:	**15mA Typical, 25mA Max**
Input Current (Disabled Mode):	**1.5mA Max**
Output:	**HCMOS**
Symmetry:	**45/55% Max @ 50%Vcc**
Rise/Fall Time:	**3ns Max @ 20% to 80% Vcc**
Logic:	**"0" = 10% Vcc Max**
	"1" = 90% Vcc Min
Load:	**15pF**
Output Current:	**±24mA Max**
Disable Time:	**200ns Max**
Start-up Time:	**1ms Typical, 2ms Max**
Pin 1 Disable Current:	**-350µA Max**
Phase Noise:	**-100 dBc/Hz Typical, -95 dBc/Hz Max at 10Hz offset**
Phase Noise Floor:	**-169 dBc/Hz Typical, -165 dBc/Hz Max**
Sub-harmonics:	**None**
Aging:	**<3ppm 1st year, <1ppm thereafter**
CCHD-957 Options:	
Temperature Range:	**0°C to +70°C (±20ppm, ±25ppm, ±50ppm)**
	-20°C to +70°C (±25ppm, ±50ppm)
	-40°C to +85°C (±25ppm, ±50ppm)

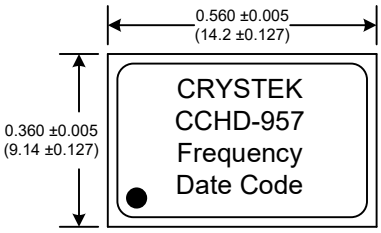
Part Number Example:
CCHD-957X-25-49.152 = 3.3V, 45/55, -40°C to +85°C (±25ppm), 49.152 MHz

Mechanical:	
Shock:	MIL-STD-883, Method 2002, Condition B
Solderability:	MIL-STD-883, Method 2003
Vibration:	MIL-STD-883, Method 2007, Condition A
Solvent Resistance:	MIL-STD-202, Method 215
Resistance to Soldering Heat:	MIL-STD-202, Method 210, Condition I or J
Environmental:	
Thermal Shock:	MIL-STD-883, Method 1011, Condition A
Moisture Resistance:	MIL-STD-883, Method 1004

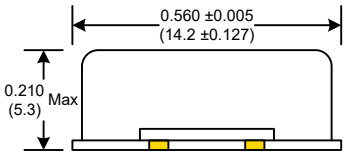
Developed Frequencies		
10.000 MHz	24.576 MHz	40.000 MHz
20.000MHz	25.000 MHz	45.1584 MHz
22.5792 MHz	27.000 MHz	49.152 MHz
24.000 MHz	28.000 MHz	50.000 MHz



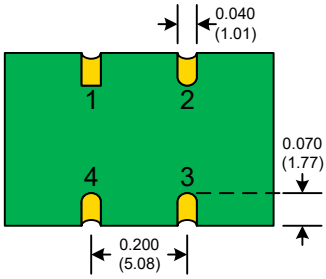
Top View



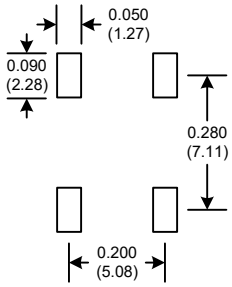
Side View



Bottom View



SUGGESTED PAD LAYOUT



RECOMMENDED REFLOW SOLDERING PROFILE
900034 (See App Note listed on website)
http://www.crystek.com/specification/reflow/900034.pdf

Tri-State/Standby Function	
Function pin 1	Output pin
Open	Active
"1" level 0.7×Vcc Min	Active
"0" level 0.3×Vcc Max	High Z

Pad	Connection
1	E/D
2	GND
3	OUT
4	Vcc

PAD FINISH: Immersion Gold (ENIG); 5 micro inches maximum

Crystek Corporation reserves the right to make changes to its products and/or information contained herein without notice. No liability is assumed as a result of its use or application.

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